

# When Coverage Diverges from Politics: Source Selection in Climate News\*

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## Abstract

How does news coverage track elite political debate? We examine whether mainstream newspaper coverage of climate change reflects the terms of that debate or filters them. Using a large-language-model pipeline, we extract 176,435 sources from 30,988 climate articles across six major U.S. newspapers (2012–2022) and classify them, alongside 26,868 congressional speeches (1994–2023), along two dimensions: scientific urgency–skepticism and economic cost–benefit. We document a widening politics–coverage gap after 2016. In Congress, scientific and economic arguments remain similarly prevalent, whereas mainstream newspapers increasingly emphasize scientific urgency while sharply reducing attention to the economic consequences of climate policy. By contrast, climate skepticism declines in parallel across both Congress and the press. Linking articles to career histories for 3,688 journalists, we show that this divergence is driven primarily by changes in journalist composition. These findings demonstrate how source selection and newsroom composition shape the political information citizens receive.

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# 1 Introduction

The news media shape policy narratives through the sources they cite (Sigal, 1986). Which voices journalists include—and exclude—affects what readers learn and how they evaluate policy trade-offs (Shapiro, 2016). Consider coverage of the Biden administration’s April 2024 Environmental Protection Agency (EPA) rule imposing stricter pollution controls on existing coal-fired power plants. Both the *New York Times* (NYT) and the *Wall Street Journal* (WSJ) reported on the policy (Friedman and Davenport, 2024; Hiller and Niiler, 2024). Each quoted EPA Administrator Michael Regan who announced the rule, as well as Mac McLennan, CEO of Minnkota Power Cooperative, who expressed concerns about the rule’s timeline and feasibility. The NYT, however, also cited Harold Wimmer, president of the American Lung Association, emphasizing the regulation’s public health benefits. By contrast, the WSJ quoted Daniel Cohan, an associate professor of civil and environmental engineering at Rice University, to highlight the financial challenges facing coal-fired power plants and the regulatory costs to firms. Although both outlets covered the same policy, differences in source selection produced distinct narratives of its consequences.

Quoted sources constitute a substantial share of news content and play a central role in constructing narrative frames. Although prior work has focused on identifying bias in news story selection (e.g., Ansolabehere, Snowberg, and Snyder (2005); Hassell, Holbein, and Miles (2020)), less attention has been paid to how variation in sourcing shapes the information environment. The types of sources cited—and the perspectives they represent—affect both the framing of policy debates and the information conveyed to audiences. Eliaz and Spiegler (2020) formalize a framework in which political disagreement arises from competing narratives about underlying reality. Within this framework, media outlets serve as key intermediaries, and we argue that source selection is a primary mechanism through which they construct and transmit these narratives.

In theory, the objectivity norm that governed twentieth-century American journalism encouraged balanced sourcing across competing perspectives (Tuchman, 1972; Schudson, 2001). Climate change has become the principal setting in which scholars have examined the operation—and lim-

its—of this norm. Because journalists have long faced the challenge of covering an issue characterized by a strong scientific consensus alongside organized political and economic interests advancing competing narratives (Oreskes and Conway, 2010), climate reporting has served as a central case for debates over whether balanced sourcing enhances or distorts public understanding (Boykoff and Boykoff, 2004, 2007; Shapiro, 2016). In practice, this norm has come under increasing strain from two forces: growing scientific consensus on climate change, which renders “both-sides” framing potentially misleading, and rising partisan polarization over climate policy (Karol, 2019; Egan and Mullin, 2023), which has made source selection itself ideologically contested. Most existing work on this shift examines whether media coverage of climate science has become less balanced between mainstream scientists and skeptics (e.g., Dunlap, McCright, and Yarosh (2016); Wetts (2020)). This question—typically framed along an urgency–skepticism axis—is largely resolved: overt climate denial has declined substantially in major newspapers (Schmid-Petri et al., 2017).

Yet this is not the dimension along which public opinion is most divided. Americans broadly accept that climate change is occurring: as of 2025, an estimated 69% of U.S. adults think global warming is happening, a clear national majority<sup>1</sup>; disagreement instead centers on the economic costs of climate policy, including regulatory burdens, employment effects, and taxes (Egan and Mullin, 2017; Painter, 2023). This economic dimension now constitutes the core of Republican climate rhetoric in Congress and presents a subtler challenge to journalistic norms. Unlike outright skepticism, cost-based opposition to regulation is not scientifically disqualifying, and presenting such views alongside pro-climate arguments does not constitute false balance.

Survey evidence from the Cooperative Election Study (2008–2024) shows how these competing narratives align with mass opinion. Throughout the period, Democratic respondents consistently express much stronger support for prioritizing environmental protection over jobs than Republican respondents. Since 2015, roughly 85–90% of Democrats have favored environmental protection, compared with only 40–45% of Republicans, yielding a persistent partisan gap.

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<sup>1</sup><https://climatecommunication.yale.edu/visualizations-data/ycom-us/> (accessed July 10, 2026).

This enduring divide mirrors elite polarization over climate policy (Karol, 2019) and suggests that contemporary political conflict centers less on the scientific reality of climate change than on the economic trade-offs associated with environmental regulation.

Recent research argues that news coverage largely reflects rather than leads public opinion. Journalists respond to audience preferences, policy priorities, and public concerns, causing news coverage to track the issues citizens care about and the positions they hold (Wlezien, 2024; Wlezien and Soroka, 2024; Hassell and Wlezien, 2025). If so, mainstream news coverage should continue to represent the competing economic arguments surrounding climate policy, where both elite debate and public opinion remain deeply divided. We examine whether it does. Addressing this question requires three elements largely absent from prior work: large-scale data on source-level citations; a framework that distinguishes between the scientific and economic dimensions of climate disagreement; and a benchmark capturing what political actors actually say, rather than what appears in the press. We provide all three.

First, we compile a corpus of climate policy–related news articles from six major U.S. newspapers—the *New York Times*, *Washington Post*, *Wall Street Journal*, *Los Angeles Times*, *Chicago Tribune*, and *USA Today*—published between 2012 and 2022. From each article, we identify and classify cited sources. This task is nontrivial because source attribution is often embedded in complex textual structures, with variation in names, pronouns, and contextual references. To address this challenge, we develop a method that leverages large language models to systematically and cost-effectively extract source mentions from article text. Applying this approach, we identify 176,435 cited sources across 30,988 articles. Validation against crowdworker-corrected samples indicates error rates below 5% for false positives, omissions, and incorrect source details.

Second, we classify each cited source along two independent dimensions. The urgency–skeptic axis captures disagreement over climate science: urgency sources emphasize that climate change poses a pressing threat, whereas skeptic sources contest the scientific consensus. The economic-benefit/economic-cost axis captures disagreement over the costs and benefits of climate policy: economic-cost sources emphasize regulatory burdens, job losses, or higher energy prices, while

economic-benefit sources highlight employment gains in renewable energy or long-run cost reductions. These dimensions are conceptually distinct. A source may accept the scientific consensus while opposing regulation on economic grounds, or support renewable energy for economic reasons without invoking climate risk. This distinction constitutes the paper's central analytic contribution. It allows us to separate the decline of false balance on the scientific dimension from selective filtering on the economic dimension, which has different empirical patterns and normative implications.

Third, assessing whether coverage filters political speech requires an independent measure of that speech. We therefore construct a parallel corpus of 26,868 congressional speeches on climate policy from 1994 to 2023 and classify each speech along the same two dimensions. Congress provides a natural benchmark: it is the primary arena for national climate policy debate, politicians are among the most frequently cited sources in news coverage, and the distribution of arguments by party captures the supply of claims available for journalistic coverage.

Leveraging novel data on source-level citations and congressional speeches, we document a substantial politics–coverage gap that widens especially after 2016. Throughout the period, Congress devotes similar attention to the scientific and economic dimensions of climate policy. Mainstream newspapers increasingly diverge from this pattern by emphasizing scientific urgency while placing progressively less emphasis on the economic consequences of climate policy. By contrast, climate skepticism declines in parallel across Congress and the mainstream press, indicating that the politics–coverage gap is driven not by differences in scientific debate but by selective filtering of the economic dimension. As a comparison, we also examine Breitbart, a prominent conservative news outlet outside the mainstream press. Unlike mainstream newspapers, Breitbart increasingly cites sources emphasizing the economic consequences of climate policy after 2018, bringing its coverage substantially closer to the balance of scientific and economic arguments observed in Congress. (HY: we don't want to give an impression that Breitbart is a balanced journalism...it is possible that economic cost dimension is also a narrative that fossil fuel firms push after realizing that denial does not work.)

To understand the factors driving the coverage gap, we examine the role of individual journalists. While much of the existing literature attributes media slant to outlet-level incentives, including revenue maximization and audience demand (Gentzkow and Shapiro, 2010; Martin and Yurukoglu, 2017; Hopkins, Lelkes, and Wolken, 2005), more recent work emphasizes the influence of journalists themselves (Cagé et al., 2022; Kim, Lelkes, and McCrain, 2022). We assemble career and demographic data for 3,688 journalists from LinkedIn and MuckRack and ask whether the widening coverage gap reflects changes in reporters’ citation behavior or changes in the composition of the journalistic workforce, particularly amid recent efforts by news organizations to diversify their newsrooms.

We find that changes in journalist composition account for a substantial share of the widening gap. Within the same newspaper and year, reporters whose climate-beat debut occurred after 2016 cite 9 percentage points more urgency sources and 2 percentage points fewer economic-cost sources than earlier cohorts, corresponding to a 17% increase and a 33% decrease relative to baseline means. This cohort difference emerges abruptly in 2017—a period marked by major changes, including the inauguration of Donald Trump, the U.S. withdrawal from the Paris Agreement, and intensified efforts by news organizations to diversify their newsrooms—and persists thereafter. Using LinkedIn-based measures of career entry produces even larger estimates, while adding newspaper-section fixed effects attenuates but does not eliminate the results. We also find evidence of within-journalist change: holding reporter identity constant, journalists become less likely to cite economic-cost sources over time.

This paper makes three contributions. First, we develop a general approach to measuring news slant through the sources journalists cite. Existing measures typically locate coverage on a single liberal–conservative dimension using indicators such as think-tank citations (Groseclose and Milyo, 2005). We instead treat cited sources as the building blocks of news coverage: using a large language model, we extract and classify every quoted or paraphrased source along issue-specific dimensions of political disagreement. Applied to climate coverage, this approach distinguishes disagreement over climate science from disagreement over the economic consequences of climate

policy—a distinction that one-dimensional measures cannot capture and that proves central to our findings. More broadly, our framework provides a general approach for measuring how source selection shapes news narratives across contested policy domains, including immigration and trade.

Second, we benchmark news coverage against the supply of elite political speech. We construct a parallel corpus of congressional speeches and classify them along the same two dimensions, allowing us to compare the arguments available to journalists with those that appear in news coverage. We document a politics–coverage gap specific to the economic dimension: while coverage mirrors the decline of climate skepticism in Congress, it increasingly filters out economic arguments against climate policy that remain prevalent in elite political debate. Because public opinion remains sharply divided over these economic trade-offs, this divergence may contribute to growing distrust in the news media by reinforcing perceptions that journalists selectively represent one side of the policy debate (Ladd, 2012; Brennan, 2025; Kates, Ladd, and Tucker, 2026; Zheng, 2026), even when their reporting reflects the scientific consensus. Such perceptions may also amplify the effectiveness of political elites’ attacks on mainstream news organizations, which can reduce citizens’ trust in information providers and their willingness to engage with and be persuaded by news coverage (Peterson et al., 2026).

Third, we bring individual journalists to the center of the analysis (Baron, 2006). While recent work documents substantial ideological variation within news outlets (Braghieri et al., 2025; Green et al., 2025), relatively little is known about who generates that variation. Linking articles to reporters’ career and demographic information, we show that differences in sourcing are substantially larger across journalists than across outlets, and that journalist replacement accounts for most of the observed change. These findings suggest that entry into the profession—who becomes a reporter—plays a central role in shaping the information available to the public.

## 2 Source Selection and the Representation of Political Debate

News stories are built around sources. Quotations and paraphrases from politicians, bureaucrats, experts, businesses, and advocacy organizations provide much of the factual content and interpretive context of news coverage. As [Sigal \(1986\)](#) famously argues, “news is, after all, not what journalists think, but what their sources say.” Because journalists exercise considerable discretion over whom they quote, source selection shapes the perspectives readers encounter and the narratives constructed around political events ([Whitney et al., 1989](#); [Baron, 2006](#); [Carlson, 2009](#)). Differences in source selection can therefore produce markedly different interpretations of the same event even when the underlying facts remain unchanged ([Eliaz and Spiegler, 2020](#)).

Source selection is also one of the principal mechanisms through which media slant is produced. By choosing which actors are allowed to define a policy problem, explain its causes, or evaluate its consequences, journalists determine which arguments become visible to readers. This insight underlies an existing literature measuring media slant using cited sources. For example, [Groseclose and Milyo \(2005\)](#) estimate ideological differences across newspapers by comparing their citations of think tanks to those cited by members of Congress. Our paper builds on this insight but asks a different question: not whether sources are ideologically balanced, but whether the distribution of sources reflects the distribution of arguments in contemporary political debate.

[\[HY: should we discuss Wlezien et al.’s papers on the news and the reflection of public opinions here?\]](#)

Historically, this discretion was constrained by the objectivity norm, which encouraged journalists to present competing perspectives on controversial political issues ([Tuchman, 1972](#); [Schudson, 2001](#)). Climate change has become one of the clearest challenges to this norm. Early scholarship argued that balanced reporting systematically overrepresented climate skeptics relative to the overwhelming scientific consensus, creating “false balance” in public discourse ([Boykoff and Boykoff, 2004, 2007](#); [Shapiro, 2016](#)). More recent evidence suggests that this concern has diminished as overt climate skepticism has largely disappeared from mainstream newspapers ([Schmid-Petri et al.,](#)



2017). Accordingly, much of the literature evaluates climate coverage by asking whether journalists have moved away from balancing scientific consensus against denial.

This focus on scientific disagreement, however, no longer captures the principal dimension of political conflict over climate policy. While most Americans accept that climate change is occurring, partisan disagreement increasingly centers on the economic consequences of climate policy, including regulatory costs, employment effects, and energy prices (Egan and Mullin, 2017; Painter, 2023). To characterize this dimension of public opinion, we draw on the Cooperative Election Study (CES), which has repeatedly asked respondents to evaluate the trade-off between environmental protection and economic costs. From 2008 to 2012, respondents were asked whether protecting the environment was more important than protecting jobs and living standards. Beginning in 2015, the survey instead asked whether respondents supported environmental regulations despite their potential employment costs, such as strengthening EPA enforcement of the Clean Air Act even if doing so costs U.S. jobs.<sup>2</sup> Figure 1 plots responses from 2008 to 2024. Democrats consistently prioritize environmental protection over jobs, whereas Republicans are substantially less likely to do so. Since 2015, the partisan gap has remained stable at roughly 40–45 percentage points.

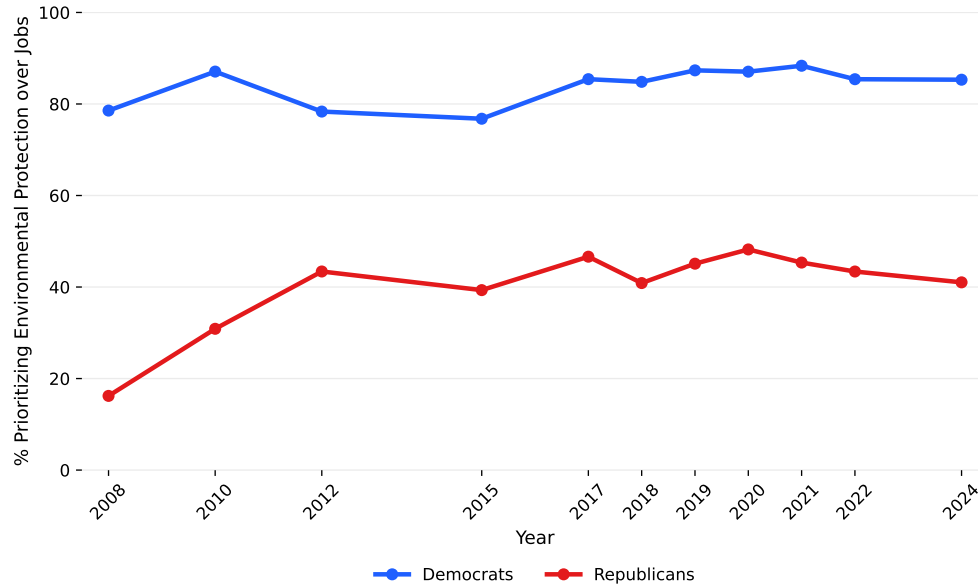
Disagreement over the trade-off between environmental protection and economic costs remains central to American climate politics. Unlike scientific skepticism, however, economic objections are not factually incompatible with the scientific consensus. Consequently, excluding such arguments cannot be justified as avoiding false balance. Existing work has largely focused on whether mainstream newspapers overrepresent scientifically unsupported claims. We argue that an equally important question is whether newspapers accurately represent the economic dimension of contemporary political conflict over climate policy. Put differently, do mainstream newspapers reflect the distribution of arguments that characterize political debate, or do they selectively filter one dimension of that debate through their source selection?

Answering this question requires measuring both the arguments supplied by political actors

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<sup>2</sup>Question wording varies across survey waves. Appendix G provides the exact wording and coding procedures.

Figure 1: Partisan Preferences over Environmental Protection versus Job Protection, 2008–2024



*Notes:* The figure reports the share of respondents prioritizing environmental protection over jobs, by political ideology. The survey wording changes slightly after 2013 but consistently measures support for environmental protection despite potential economic costs.

and the arguments that ultimately appear in news coverage. We therefore compare the distribution of climate arguments in Congress with the arguments represented through sources cited in mainstream newspapers. Congress provides a natural benchmark because it is the primary forum in which elected officials publicly articulate competing arguments over national climate policy. Because members also have strong electoral incentives to advance arguments that resonate with their constituents, congressional debate provides a useful benchmark for the broader political cleavages surrounding climate policy. By examining both the scientific and economic dimensions of climate debate, we distinguish a decline in false balance from selective filtering of economically based arguments. The next section introduces the datasets and measurement strategy we develop to do so.

### 3 Data and Methods

Our empirical analysis combines four sources of data. First, we collect congressional speeches to measure the supply of political arguments surrounding climate policy. Second, we compile a corpus of climate news articles and identify the cited sources in each article using a large-language-model pipeline. Third, we classify both congressional speeches and news sources along two dimensions of climate disagreement, allowing direct comparison between political debate and media coverage. Finally, we assemble career and demographic information on journalists to examine the mechanisms underlying any observed divergence.

#### Congressional Speech Data

We construct a corpus of congressional climate speeches from two sources: the Gentzkow-Shapiro Congressional Record corpus covering the 103rd–114th Congresses (1994–2016) and the Congressional Record API for the 115th–118th Congresses (2017–2023). We clean the corpus by dropping procedural, nonsubstantive speeches and then identify climate-relevant speeches using a combination of keyword filtering and large language model based classification. In total, we extract 25,749 climate speeches (about 2% of all total speeches). Full details on our extraction, cleaning, and filtering procedure can be found in Appendix A.

Figure A1 shows the number of climate speeches made by each party over time since 1994. The volume of climate speeches begins to rise dramatically in the late 2000s which is likely driven by the 2007 release of the IPCC Fourth Assessment Report which a major landmark in the recognition of global climate change as both a human powered phenomenon and existential threat (IPCC Core Writing Team, Pachauri, and Reisinger, 2007). Congressional Democrats consistently make more climate related speeches than Republicans (543 vs. 324 per year on average).

## Newspaper Articles and Sources

We collect all climate and environment related news articles from six major U.S. newspapers—the *New York Times*, *Washington Post*, *Wall Street Journal*, *Los Angeles Times*, *Chicago Tribune*, and *USA Today*—published between 2012 and 2022. We supplement these mainstream outlets with a comparable corpus scraped from *Breitbart News*, which covers the same period and allows us to compare how a partisan conservative outlet represents the same political debate. In total, we collect 30,988 articles.<sup>3</sup> We note a steady increase in the absolute number of climate articles over time across most outlets, with a notable surge in *The New York Times* and *The Washington Post* around 2018–2021 (Figure A2).

We then identify the cited sources in each article. Identifying these sources is challenging because of the nuanced integration of source information within the article text; the same source may be referenced multiple times in different ways using variations in name, pronouns and context for each source. We develop a method that leverages the text processing capabilities of large language models to reliably and cost-effectively identify source citations from each article in our dataset. We divide this task into subtasks: extracting attribution spans, aggregating these spans to speaker/organization, and converting to structured JSON. The details of the method are shown in Figure A3. For each source citation, we identify the name of the source (where applicable), their organizational affiliation, and a brief summary of their contribution in the context of the news article. We extract 176,435 sources in total (median 6 per article, SD = 2.8). Validation using crowdworkers on 214 sources across 50 articles finds error rates below 5%.<sup>4</sup>

We additionally categorize sources based on their institutional roles, dividing them into six general types: politicians, bureaucrats, academic researchers, advocacy groups, businesses, and media organizations (plus an “other” category). We further identify party affiliations for U.S. politicians and distinguish organizations tied to fossil fuel interests or environmental advocacy. Table 1 presents the distribution of source types by newspaper. Politicians and bureaucrats make

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<sup>3</sup>Details on article collection and filtering in Appendix A.

<sup>4</sup>Appendix B provides full validation details.

up the largest share— about one-quarter — although academic sources (17%) and advocacy groups (13%) also account for substantial shares.

Table 1: Source Categories and Distribution

| Category               | %       | Description                                      |
|------------------------|---------|--------------------------------------------------|
| Politician             | 13.0%   | Governors, legislators, White House spokespeople |
| Republicans            | 3.2%    |                                                  |
| Democrats              | 8.7%    |                                                  |
| Other/Unknown          | 1.0%    |                                                  |
| Bureaucrat             | 12.2%   | EPA, NOAA, NASA                                  |
| International          | 9.9%    | UN, IPCC                                         |
| Environmental Advocacy | 10.0%   | WWF, Sierra Club, Greenpeace                     |
| Fossil Fuel Interests  | 3.0%    | API, ExxonMobil, BP                              |
| Other Advocacy         | 2.8%    | Nonprofits, think tanks, activists               |
| Other Business         | 1.9%    | BlackRock, Amazon                                |
| Academic               | 17.0%   | Academics and government labs                    |
| Media                  | 8.7%    | Newspapers, media outlets                        |
| Other                  | 21.5%   | Lawyers, private citizens, others                |
| Total                  | 176,435 |                                                  |

## Two Dimensions of Climate Disagreement

Our primary analytical contribution is characterizing what argument each source makes in a two dimensional space of climate discourse. We contend that a one-dimensional pro/anti classification obscures meaningful variation in climate politics. One may acknowledge the reality of climate change while opposing regulation because of economic concerns; similarly, one may support renewable energy because of economic opportunities while placing little emphasis on climate benefits. To capture these distinctions, we classify each cited source along two independent dimensions: an urgency-skeptic axis and an economic cost-benefit axis.

The urgency-skeptic axis captures disagreement over the scientific foundation of climate change and the necessity of immediate action. At one end of the spectrum, “urgency” arguments empha-

size that climate change poses a serious and immediate threat requiring substantial intervention. On the other, “skeptical” arguments question the scientific consensus, minimize the role of human activity, or argue that climate impacts are overstated. The economic cost-benefit axis captures disagreement over the economic consequences of climate policy. Economic-cost arguments emphasize potential costs to industries, workers, consumers, or government budgets as reasons to delay or limit climate action. Economic-benefit arguments emphasize economic benefits associated with climate policy, such as employment growth or technological opportunities. Importantly, these dimensions are analytically distinct: one can simultaneously accept the scientific basis of climate change and oppose particular policies because of their perceived economic costs.

We label each news citation in our dataset as either primary using urgency, skeptic, economic-cost, or economic-benefit based arguments using gpt-4.1. For example, a quotation from Jagoda Egeland of the OECD International Transport Forum describing aviation industry emissions as “a big crisis, a pressing crisis ” is coded as “urgency,” which is overall the most commonly used argument in our data (40.3%). Meanwhile, economic cost and benefit arguments are relatively rare (4.2% each), and arguments for climate skepticism are rarer still (2%). 49% of sources do not use any of these four arguments; 26% contain factual references to policy statements or proposals, 12% reference industry commitments and actions, and 11% are miscellaneous other references.<sup>5</sup> Additional details on our codebook, labeling scheme, and other source types are available in the Appendix.

We describe the kinds of arguments made by different types of sources in Figure 2. We aggregate citations to the level of the organization or individual politician. For each source (e.g., Mitch McConnell or the World Wildlife Foundation), we calculate the relative balance of urgency versus skeptic arguments and economic-benefit versus economic-cost arguments. Sources that make no arguments on a given dimension are assigned a value of zero.

For clarity, the left panel of the figure restricts attention to politicians, fossil fuel interests, and environmental advocacy groups. These sources separate cleanly along both dimensions. Repub-

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<sup>5</sup>Typically, the catch-all other bucket contains statements from actors, religious leaders, activists, and other groups supporting or opposing climate action but without invoking our primary frame focuses.

lican politicians and fossil fuel interests tend to emphasize skeptic over urgency arguments and economic-cost over economic-benefit arguments. In contrast, Democratic politicians and environmental advocacy groups cluster in the urgency/economic-benefit quadrant. However, the two dimensions are not reducible to a single ideological axis. Several politicians and organizations simultaneously acknowledge the urgency of climate change while emphasizing its economic costs. Meanwhile, the right panel shows that sources without explicit ideological affiliations express a wide variety of arguments on the economic dimension while predominantly making urgency arguments on the urgency-skeptic dimension.

These patterns illustrate the value of treating the climate debate as two distinct dimensions rather than a single ideological continuum. To compare our approach with a conventional one-dimensional measure, we map sources into an ideological space using DIME scores (Appendix ??). The resulting patterns are broadly similar: left-leaning sources generally emphasize urgency and economic-benefit arguments, but the two-dimensional framework also reveals substantial within-ideology variation that is obscured by a single ideological scale. Consistent with this, the correlations between DIME ideology and the urgency-skeptic and economic-cost-benefit dimensions are modest, approximately 0.3.

We also label each congressional speech using the same conceptual framework and general method: each speech is labeled as focused on urgency, skepticism, economic-benefit, or economic-cost. 81% of congressional climate speeches are focused on one of these four climate arguments. In section 4, we compare the narratives of sources cited in news to those most often found in congressional speech to assess the presence of a politics-coverage gap in climate articles.

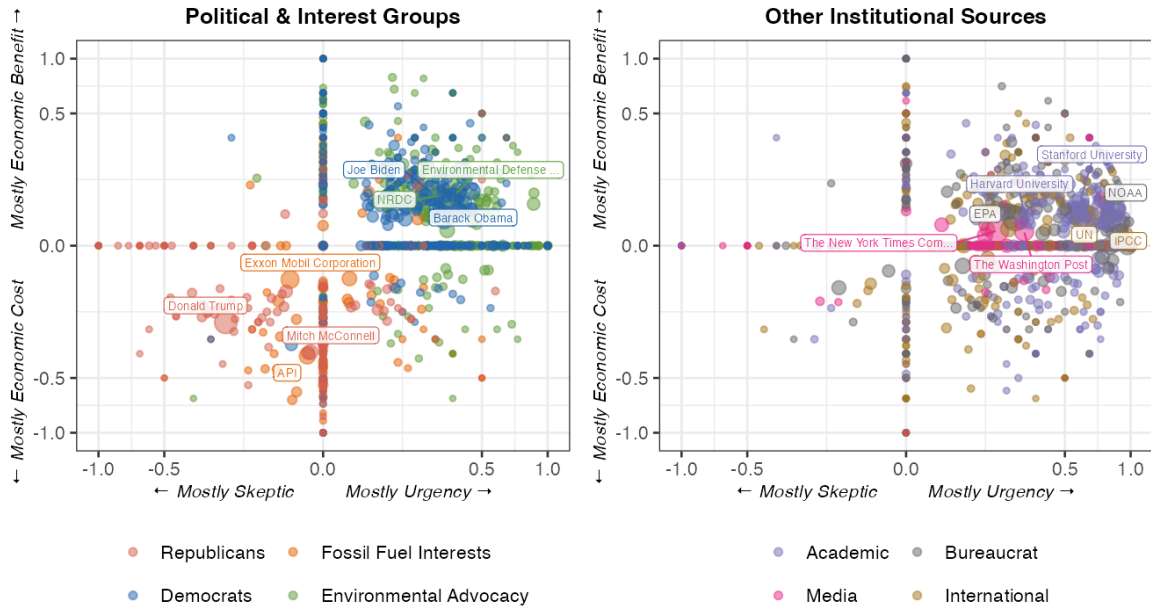
## **Journalist Data**

We collect career and demographic information for 3,688 journalists in our news sample using LinkedIn, MuckRack, and supplementary manual data collection.<sup>6</sup> For each journalist, we retrieve undergraduate graduation year, variables for education (Ivy League/selective institution, postgrad-

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<sup>6</sup>Appendix D provides details on the matching procedure and summary statistics.

Figure 2: Climate Arguments by Source Type



*Notes:* Each point represents an organization or individual politician. Axes show the difference in argument shares as a proportion of total citations:  $(urgency - skeptic) / N$  on the x-axis and  $(economic-benefit - economic-cost) / N$  on the y-axis, where  $N$  is the source's total citation count. Axes are square-root scaled to spread apart sources clustered near zero. Point size reflects total citation count per source. Labels show the two most-cited sources for each source type.

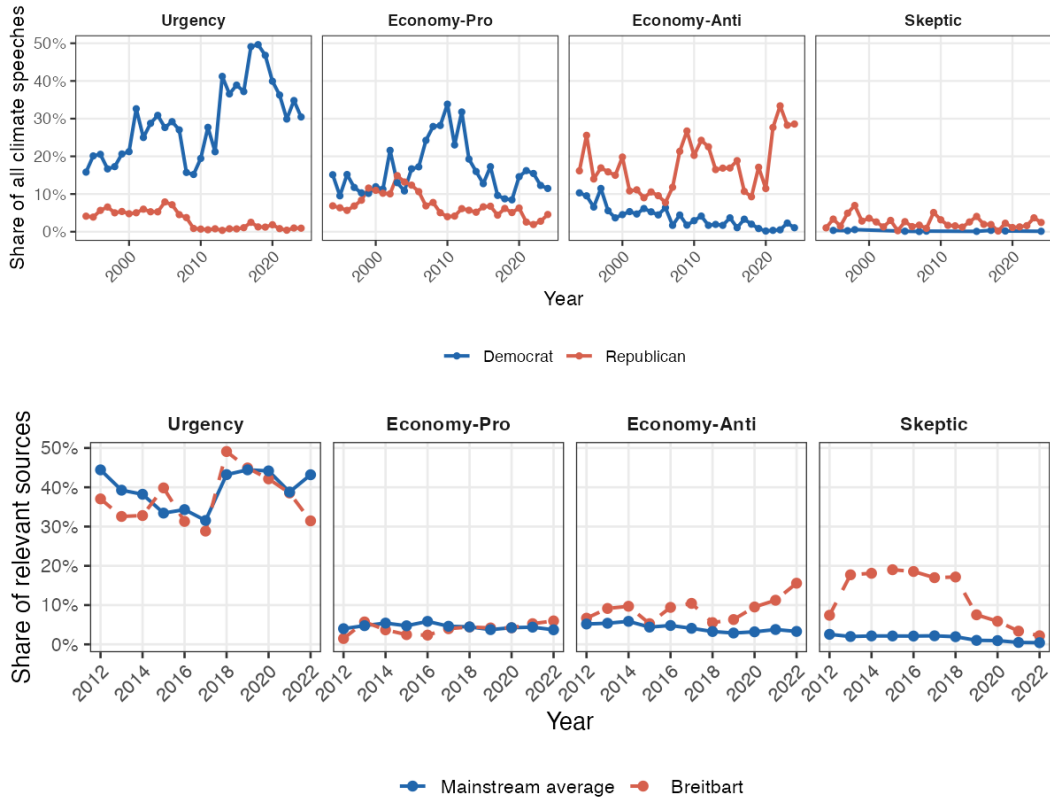
uate degree, journalism training), gender (predicted from name), race (predicted from name), and the first year for which LinkedIn reports a newsroom related work experience.

## 4 Politics-Coverage Gap in Climate Articles

Figure 3 shows the share of each type of climate argument across climate speeches and news source citations. The top panel displays the share of climate speeches in each frame category by party in Congress, 1994–2023. economic-cost is the dominant Republican frame throughout the period; starting in the late 2000s over 30% of climate speeches emphasize economic costs, regulatory burdens, and job losses associated with climate policy. Meanwhile, although economic-benefit framing enjoyed a relative peak in the late 2000s (during the 2008-2009 cap-and-trade debate), urgency has supplanted economic framing as the dominant Democratic frame since the



Figure 3: Climate Framing Over Time: Congress and Newspapers



*Notes:* **Top panel:** Share of all climate speeches in each frame category by party in Congress. Hein-Daily corpus covers 1994–2016; Congressional Record API covers 2017–2023. **Bottom panel:** Share of climate sources in each frame category among six mainstream newspapers and Breitbart News, 2012–2022.

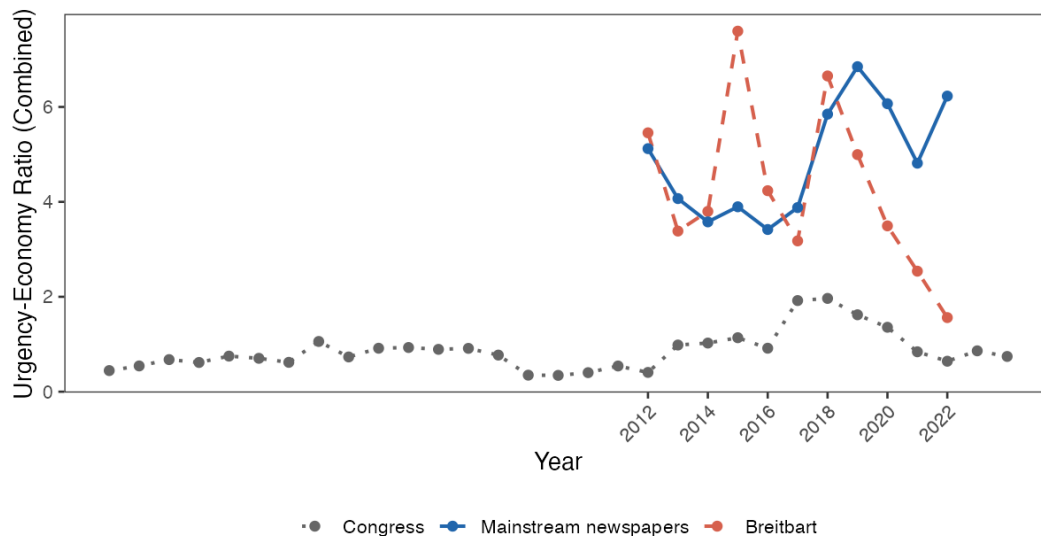
early 2010s. Climate skepticism, while more prevalent among Republican speechmakers than Democratic ones, composes a consistently small share of congressional arguments.

The bottom panel shows framing shares among climate-related sources cited in the six mainstream newspapers along with Breitbart News, 2012–2022. Urgency is the largest substantive frame in mainstream newspaper sources, enjoying over 40% of citations. All other frames, including both economic-cost, economic-benefit, and skeptic are relatively rare in mainstream news, in combination totaling to less than climate related 10% of source shares.<sup>7</sup> Breitbart news features substantively both more skeptic and economic-cost frames compared to mainstream news, although the share of skeptic frames has substantively decreased since 2018.

<sup>7</sup>Economically oriented sources are more prevalent in the Wall Street Journal compared to other outlets, but reaches 10% at most.

The contrast between the two panels of Figure 3 reveals a systematic supply-demand gap. While urgency frames are popular in Democratic speech, it dominates news coverage. However, although economic frames are also popular among Democrats and Republicans alike, economic frames receive strikingly little attention in mainstream news sources. Only Breitbart coverage cites a substantive proportion of economic-cost sources, although this share still falls short of the relative proportion in Congressional speech.

Figure 4: Urgency / Economy Ratio: Congress, Mainstream Newspapers, and Breitbart



*Notes:* Ratio of urgency (plus skeptic) to economy (both anti and pro) contributions by year. For Congress, counts of climate speeches; for newspapers, counts of cited sources. Values above one indicate urgency predominates.

To illustrate the general relationship of the urgency debate versus the economic debate, Figure 4 plots the ratio of urgency dimension to economy dimension framings over time for Congress, mainstream newspapers, and Breitbart. This ratio indicates the relative proportion of urgency and skeptic sources and speeches on the one hand to both economy anti and pro sources and speeches on the other. In Congress, the ratio hovers close to one throughout the entire 1994-2023 period—one urgency oriented speech for every economy oriented speech. Mainstream newspapers, by contrast, cite dramatically more urgency sources relative to economy sources over time, especially after 2018. By 2018-2023, urgency sources outnumber economic-cost sources by almost a factor of six. Meanwhile, Breitbart cites more urgency sources than Congress especially during 2012-2018,

but by 2018-2023 Breitbart’s focus on economic issues brings its ratio much closer to Congress’. These patterns suggest that mainstream newspapers do not simply mirror the partisan composition of political speech, but selectively filter the economic counter-framing that constitutes the core of Republican climate rhetoric.

## 5 What Drives the Gap? Journalist Composition

Our results suggest that the news media plays a role in filtering climate coverage: economic arguments (in plentiful supply by political actors) are markedly less likely to appear in mainstream newspaper sources compared to urgency arguments. We explore one potential explanation: changes in journalist composition. Journalists are relatively understudied as political agents (Baron, 2006; Hassell, Miles, and Morecraft, 2023); however a growing literature suggests that they are able to exert agency in slanting coverage toward their ideological preferences (Cagé et al., 2022; Boxell and Conway, 2022). While newsrooms historically maintained a standard of neutral objectivity (Schudson, 1978), recent survey data suggests that newer journalists are more likely to view the newsroom’s role in politics as that of an active participant rather than a neutral observer (Forman-Katz and Jurkowitz, 2022). At the same time, news organizations have become increasingly differentiated in their ideological orientations and professional norms, creating opportunities for journalists to select into outlets whose editorial missions align with their own commitments (Hagar, Wachs, and Horvát, 2023; Hassell, Miles, and Morecraft, 2023). If these changes influence sourcing decisions, newer cohorts entering mainstream news organizations should place greater emphasis on climate urgency, while newer cohorts entering conservative outlets should increasingly emphasize economic and skeptical frames.

To assess this question, we compare citation patterns between journalists with more and less newsroom experience and between mainstream news outlets and Breitbart. We measure journalist experience by assigning each journalist to a cohort based on their first year of observed climate coverage. Breitbart provides a useful comparison because it occupies a distinct ideological niche

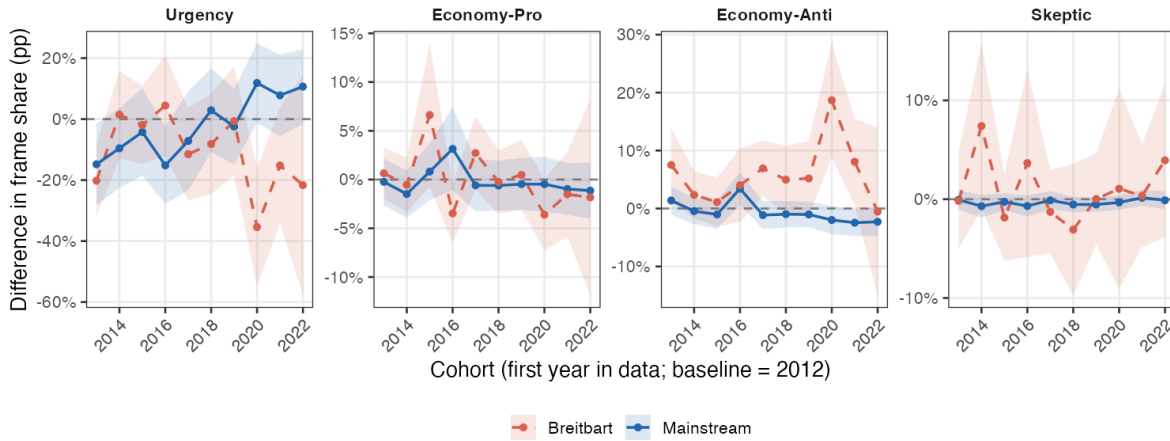
in the media ecosystem. If cohort effects simply reflect generational changes common to all journalists, we should observe similar patterns across outlets. By contrast, if they reflect ideological sorting into organizations with different editorial missions, cohort effects should diverge between mainstream newspapers and Breitbart.

We estimate a series of OLS regressions

$$Y_i = \beta_0 + \gamma_k + \delta_t + \lambda_s + \varepsilon_i \quad (1)$$

where  $Y_i$  is an article-level outcome—the share of economic-cost, urgency, economic-benefit, or skeptic sources— $\gamma_k$  is a cohort (entry-year) fixed effect,  $\delta_t$  is a year fixed effect, and  $\lambda_s$  is a newspaper-level fixed effect. Standard errors are clustered at the journalist level. Our fixed effects strategy ensures that journalist comparisons are made within the same year and newspaper, eliminating the influence of common time trends and outlet-level differences in assignment (such as editorial influence).

Figure 5: Cohort Fixed Effects on Framing: Mainstream vs. Breitbart



*Notes:* Mainstream: pooled cohort fixed effects, year and outlet FE. Breitbart: cohort fixed effects, year FE only. Journalist-clustered 95% CIs.

Figure 5 presents cohort fixed effects from these regressions separately for mainstream and Breitbart journalists.<sup>8</sup> The results are consistent with ideological sorting. Mainstream journalists

<sup>8</sup>Journalists in all mainstream outlets generally behave similarly to each other; see Appendix for results by indi-

entering in later years cite substantially more urgency sources than earlier cohorts working in the same newspaper during the same year. Mainstream journalists with less experience are also less likely to cite both economic-benefit and economic-cost sources, although these effects are smaller in magnitude compared to the effects on urgency source shares. By contrast, less experienced journalists at Breitbart are less likely to cite urgency arguments and more likely to cite economic-cost arguments. The opposing cohort gradients between mainstream outlets and Breitbart suggest that the results are not simply capturing generational changes among journalists, but instead reflect differences in the types of journalists entering ideologically distinct media organizations.

To summarize these trends, we estimate an additional set of models that regress the same outcome variables on the cohort year as a simple linear variable with the same fixed effects and controls. We also include an interaction term with a dummy variable for Breitbart to model differences between mainstream outlets and Breitbart News. These models are presented in Table 2. We additionally present two robustness specifications per outcome variable. First, we present identical models that use journalists' career start dates instead of climate-reporting entry year (cohort) dates using data from LinkedIn. While we use cohort dates in our primary specification in order to best leverage the size of our sample, such cohort dates are an imperfect proxy for journalist experience; for example, journalists could switch from non-climate reporting assignments to climate related reporting assignments. For 54% of journalists in our data (N=1990), we collect the exact year they enter the field of journalism using LinkedIn data and use this variable instead of their entry year. Then, our second robustness specification includes section fixed effects in order to guard against threats from beat assignment or editor influence; Journalists with less experience may be systematically assigned to reporting on news items that are more likely to feature urgency-related sources (eg; climate reporting desk versus policy desk). In order to guard against this from confounding our results, we additionally locate the section of the newspaper each article was published in and add section fixed effects to our specification. Details on this procedure can be found in Appendix D.

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vidual news outlets.

Table 2: Cohort effects on framing — linear specification and robustness checks

| Dependent Variables:                 | Urgency           |                     |                     | Skeptic             |                                  |                     |
|--------------------------------------|-------------------|---------------------|---------------------|---------------------|----------------------------------|---------------------|
| Model:                               | Cohort<br>(1)     | Career start<br>(2) | + Section FE<br>(3) | Cohort<br>(4)       | Career start<br>(5)              | + Section FE<br>(6) |
| Cohort year                          | 0.01**<br>(0.006) |                     | 0.01**<br>(0.006)   | 0.0006*<br>(0.0004) |                                  | 0.0007*<br>(0.0004) |
| Cohort year $\times$ Breitbart       | -0.01<br>(0.01)   |                     | -0.01<br>(0.01)     | -0.02***<br>(0.006) |                                  | -0.02***<br>(0.006) |
| Career start year                    |                   | 0.004**<br>(0.002)  |                     |                     | $4.8 \times 10^{-5}$<br>(0.0001) |                     |
| Career start year $\times$ Breitbart |                   | -0.006<br>(0.004)   |                     |                     | -0.004<br>(0.004)                |                     |
| <i>Fixed-effects</i>                 |                   |                     |                     |                     |                                  |                     |
| Section                              |                   |                     | Yes                 |                     |                                  | Yes                 |
| Year + Outlet                        | Yes               | Yes                 | Yes                 | Yes                 | Yes                              | Yes                 |
| Observations                         | 26,111            | 20,488              | 23,679              | 26,111              | 20,488                           | 23,679              |

| Dependent Variables:                 | Econ-Pro          |                        |                     | Econ-Anti            |                        |                       |
|--------------------------------------|-------------------|------------------------|---------------------|----------------------|------------------------|-----------------------|
| Model:                               | Cohort<br>(1)     | Career start<br>(2)    | + Section FE<br>(3) | Cohort<br>(4)        | Career start<br>(5)    | + Section FE<br>(6)   |
| Cohort year                          | -0.001<br>(0.001) |                        | -0.001<br>(0.0009)  | -0.003***<br>(0.001) |                        | -0.003***<br>(0.0009) |
| Cohort year $\times$ Breitbart       | 0.004*<br>(0.002) |                        | 0.004**<br>(0.002)  | 0.01**<br>(0.004)    |                        | 0.01**<br>(0.004)     |
| Career start year                    |                   | -0.0010***<br>(0.0004) |                     |                      | -0.0009***<br>(0.0003) |                       |
| Career start year $\times$ Breitbart |                   | 0.002***<br>(0.0007)   |                     |                      | 0.002**<br>(0.001)     |                       |
| <i>Fixed-effects</i>                 |                   |                        |                     |                      |                        |                       |
| Section                              |                   |                        | Yes                 |                      |                        | Yes                   |
| Year + Outlet                        | Yes               | Yes                    | Yes                 | Yes                  | Yes                    | Yes                   |
| Observations                         | 26,111            | 20,488                 | 23,679              | 26,111               | 20,488                 | 23,679                |

Notes: Each set of three columns reports results for one outcome frame. Column 1: preferred specification — post-2016 debut cohort indicator, year and outlet fixed effects. Column 2: alternative cohort measure — career start year > 2016, year and outlet fixed effects. Column 3: debut cohort indicator with year, outlet, and section fixed effects (all named sections). Journalist-clustered standard errors. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Across all specifications, the general conclusions illustrated by the trends in Figure 5 hold. For mainstream journalists, less experience (whether measured by cohort year or using LinkedIn data) is associated with increased urgency citation rates and with fewer economic-benefit or economic-cost citation rates. The reverse is true for Breitbart journalists, where less experience is associated with increased rates of economy oriented source citations. The results also survive the addition of section fixed effects, suggesting that the results are not driven by systematic assignment of less experienced journalists to different reporting beats. Finally, additional robustness analysis in [Appendix E](#) also shows that these results are not an artifact of older journalists with increased seniority having expanded access to economically oriented sources.

Taken together, these findings suggest that changes in journalist cohorts contribute to the widening gap between congressional discourse and mainstream climate coverage. Rather than reflecting only uniform shifts in newsroom policy or the broader information environment, the evidence indicates that journalists entering the profession in recent years make systematically different sourcing choices, even when working alongside earlier cohorts under the same editorial conditions. Notably, this cohort break emerges after 2016, a period marked by major changes in both the political and media environments, including the election of Donald Trump and his sustained criticism of mainstream news organizations, the rollback of federal climate policies and the U.S. withdrawal from the Paris Agreement, and intensified efforts by news organizations to diversify their newsrooms.

Although our analysis cannot identify which of these developments, if any, produced the observed cohort differences, the timing suggests that the composition of journalists entering the profession changed alongside broader shifts in the political and professional context. Consistent with this interpretation, Figure A6 shows that journalists entering after 2016 are more likely to be women and members of racial and ethnic minority groups. These demographic groups are also substantially more likely in national surveys to prioritize environmental protection over jobs, suggesting one possible mechanism linking changes in newsroom composition to changes in source selection ([Need to add the CES analysis in the appendix](#)). The divergent patterns across main-

stream and conservative media are consistent with a process of ideological sorting, in which newer journalists increasingly select into outlets whose interpretive frameworks align with their views of climate politics.

## 6 Conclusion

Democratic politics is characterized by competing arguments over public policy, and the news media serve as the principal intermediary between those political debates and the public. Whether news coverage faithfully represents the distribution of arguments made by political actors, or instead selectively amplifies some while filtering others, remains a fundamental question for understanding the information citizens receive (Green et al., 2025). Using a large-scale dataset of source citations and congressional speeches on climate policy, we compare the distribution of arguments available in elite political debate with those ultimately presented in mainstream newspapers. We uncover a substantial politics–coverage gap. While newspapers closely track the disappearance of climate skepticism from congressional debate, they increasingly underrepresent economic arguments about the costs of climate policy, even though these arguments remain a central component of Republican political discourse and the principal dimension of partisan disagreement over climate policy among the public.

Our findings also identify an important source of this divergence. Linking nearly 31,000 climate articles to career histories for more than 3,600 journalists, we find that the coverage gap is driven primarily by changes in journalist composition rather than differences across news organizations. Newer cohorts of climate reporters are substantially more likely to cite urgency-focused sources and less likely to cite sources emphasizing the economic costs of climate policy, even when working alongside earlier cohorts within the same newspaper. These patterns suggest that changes in who enters the profession can meaningfully alter the information environment independently of formal editorial policy.

More broadly, the framework developed here extends well beyond climate policy. Many con-



temporary policy debates involve multiple, legitimate dimensions of political disagreement that cannot be reduced to a single ideological scale. Immigration debates, for example, often juxtapose humanitarian concerns with border security and labor-market effects; trade policy weighs consumer welfare against manufacturing employment and national security; AI regulation balances innovation against concerns about privacy, employment, and algorithmic bias; and public health policies frequently require trade-offs between disease control and economic activity. In each of these domains, quoted sources define which dimensions of the debate become visible to the public. By combining source-level citation data with independent measures of political discourse, our framework provides a general approach to assessing whether news coverage reflects the distribution of arguments in political debate or selectively filters particular perspectives.

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# Supporting Information for

## *When Coverage Diverges from Politics: Source Selection in Climate News*

### **A Data and Source Extraction Methods**

#### **A.1 Congressional Speeches**

The congressional speeches dataset combines two source corpora covering non-overlapping periods. For 1981–2016 (Congresses 97–114) we use the Hein-Daily corpus (Gentzkow, Shapiro & Taddy). For 2017–2024 (Congresses 115–118) we use the Congressional Record, downloaded via the ‘congressionalrecord’ Python package.

Speaker party is resolved differently across the two sources. For Hein-Daily, party is joined directly from the corpus’s own speaker mapping files. For the Congressional Record, each speaker turn is tagged with a bioguide ID, which we match against data from the unitedstates/congress-legislators project) to recover party and state. Both sources are then restricted to speeches by members coded as Democrat (D) or Republican (R); independents and unresolved speakers are dropped.

A speech is classified as potentially climate-related if its full text matches, case-insensitively, any of the following keyword or phrase patterns:

climate, global warming, greenhouse gas, carbon dioxide, CO<sub>2</sub>, fossil fuel, renewable energy, clean energy, cap-and-trade, carbon tax, kyoto, paris accord, paris agreement, clean air act, carbon emission.

Some floor speeches run to tens of thousands of words, typically because a member has entered a committee report or other document into the record alongside their remarks. To keep these from dominating downstream processing, any speech longer than 3,000 words is trimmed to isolate its climate-relevant portion before further use. The speech is first split into paragraphs on blank lines, falling back to sentence boundaries if it has no paragraph breaks. Paragraphs matching the climate keyword pattern are flagged, and each flagged paragraph is kept along with one paragraph of context immediately before and after it. The kept paragraphs are concatenated up to a hard cap of 2,000 words; if even the first retained paragraph exceeds that cap, it’s truncated at the word level. Speeches under the 3,000-word threshold are left unchanged. Across the pooled pre-processing sample of 22,812 speeches, 2,206 (9.7%) exceeded the threshold and were trimmed.

To express climate speeches as a share of all floor speeches, rather than just describing the climate sample on its own, we separately compute the total volume of D/R floor speeches on any topic, by year, from the same two source corpora. Raw floor-speech corpora are dominated by very short procedural remarks — “I yield back the balance of my time,” clerk readings, and motions to table. Roughly 40% of raw Hein-Daily rows are under 20 words. We therefore exclude speeches under 20 words from both the climate numerator and the all-speeches denominator before computing shares. This barely changes the climate speech count (29,094 speeches drop to 29,003 after the filter). The resulting prevalence is 2.34% overall (29,700 of 1,303,305.).

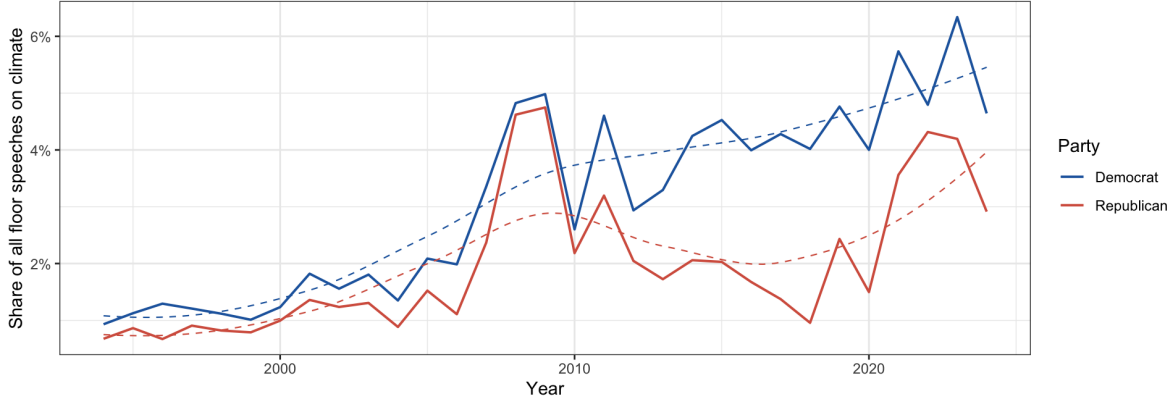


Figure A1: Number of climate speeches in each year made by members of each party.

## A.2 Identifying Climate Change Articles

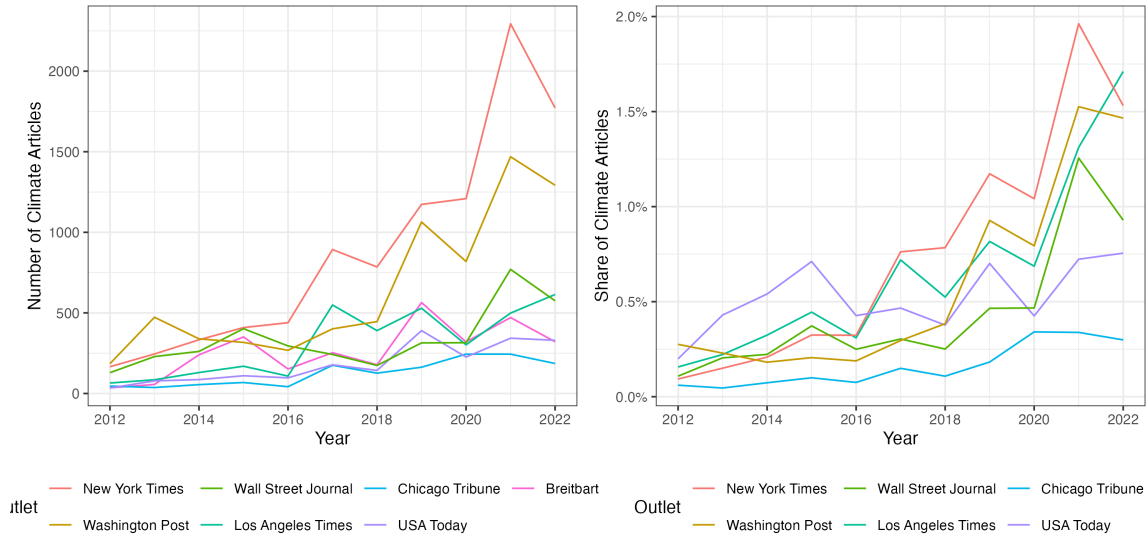
To collect mainstream news articles, we search ProQuest for all articles from six major U.S. newspapers (*New York Times*, *Washington Post*, *Wall Street Journal*, *Los Angeles Times*, *Chicago Tribune*, and *USA Today*) containing “climate change,” “environment,” or “global warming” anywhere in the text, headline, or ProQuest summary (N=86,426). We collect articles from Breitbart News, which is not available in ProQuest, directly from the Breitbart News website using a custom web scraping tool. Using this tool, we collect all Breitbart News articles tagged with Global Warming, Climate, or Environment (N=10,900).

These articles include discussion of the political strategies of candidates, announcements of staff changes at energy companies or weather forecasting agencies in addition to articles that primarily concern climate-change related topics. This set of articles contains many articles containing only a passing reference to climate change, because of the relatively permissive inclusion strategy. We identify the subset of articles that are primarily about climate change or climate change-related policy (N=32K) using the gpt-3.5-turbo large language model with the prompt “Does the following news article excerpt contain discussion of the science of climate change, global warming, impacts of climate change, or climate change related policy?” Next, we determine whether the headline and lede of each article contains substantive discussion of climate change or climate policy using gpt-4o-mini. Validation against 200 hand-labeled articles yields 95% accuracy (Precision: 97%, Recall: 93%, F1: 94.6%). In total, we retain 30,988 articles (27,372 mainstream, 3,616 Breitbart).

## A.3 Cited Source Identification and Extraction

Identifying sources cited in news articles is challenging because of the nuanced integration of source information within the article text, including variations in name, pronouns and context for each source. We develop a method that leverages the text processing capabilities of large language models to reliably and cost-effectively identify and extract the sources cited in each article in our dataset. We divide the source extraction task into manageable subtasks and develop custom prompts for each subtask. The details of the method are shown in Figure A3. First, we identify all quotes and information attributed to third party sources in each article. Second, we aggregate the quotes and information to the speaker or organization level. Finally, we transform the data into

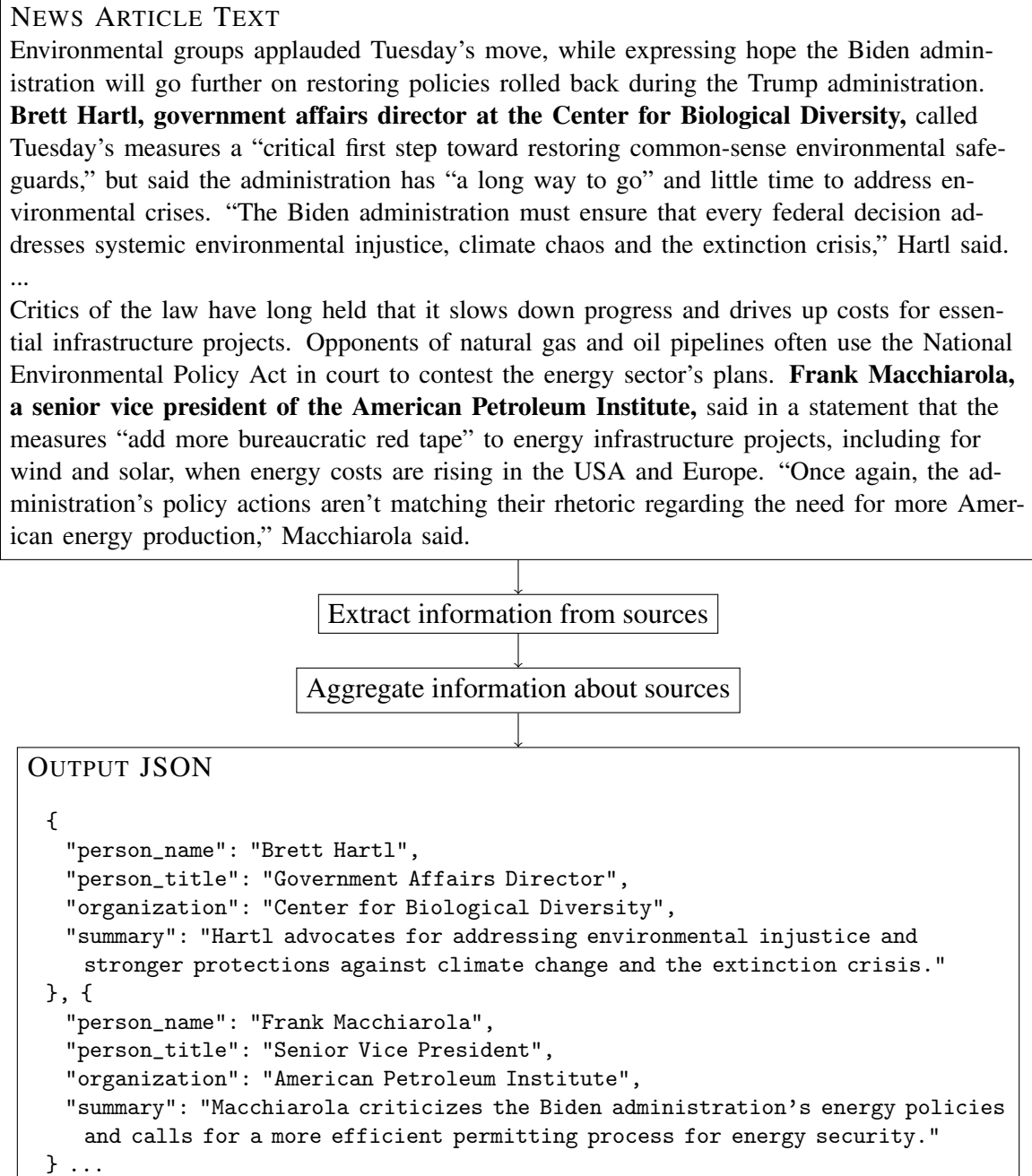
Figure A2: Number of Climate Articles by Outlet, 2012–2022



structured JSON. We employ a large set of crowdworkers and a custom validation task design to ensure the reliability of this source extraction method.

Our large-language-model based approach has the additional advantage of preserving the content of each source and the context in which it is used in each article, allowing us to enrich our dataset with further supplementary annotations without needing to re-process the original article text. Based on the source name, organization, and summary, we develop a separate large-language-model prompt to identify the category of each source. We identify eight general categories: politicians, bureaucrats, academic researchers, advocacy groups, businesses, media, and a catch-all other category. We further identify party affiliations for US politicians, and distinguish organizations specifically associated with fossil fuels or environmental advocacy. We validate our categorization technique by manually reviewing a random sample of 450 labels. Our categorization method performs with 89% accuracy on this test set. The principal source of errors stems from executive politicians who are incorrectly specified as bureaucrats and from research stemming from government affiliated institutions.

Figure A3: Source Extraction Process Outline





# B Source Extraction Validation Details

We collected corrections for 462 extracted sources for 87 articles from 25 crowd workers hired on the Prolific platform with an undergraduate or graduate degree and English fluency. Figure A4 shows a screenshot of the interface workers used to submit corrections to the extracted sources. The workers were provided the text of the article, a list of extracted sources, and details (name, title, organization, category, etc.) for each source. For each source, the workers were asked to mark the source as either (a) present in the article text, (b) present in the article text but a duplicate of another source in the list, (c) not present in the article text, or (d) unsure. The worker was also asked to correct details if there were any errors.

Figure A4: Screenshot of Tool Interface Used by Crowdworkers to Validate the Correctness of Extracted Source Citations

Climate Article SourcesHomeAnnotation CodebookStatisticsHelpFinished 0/5Current\_Id 0

go

Currently logged in as test

In Kern County, oil and gas generate tax revenues that keep schools open and police departments funded.

It's Wednesday. Kern County, which now produces 70 percent of California's oil, considers a future with no fossil fuel extraction. Plus, a dog helps rescue its owner in a California forest.

Long a leader in environmental regulation, California is considering phasing out oil drilling in the state by 2045 as part of an effort to reduce carbon emissions that contribute to global warming.

Our state's economy isn't heavily dependent on fossil fuels the way the economies of, say, West Virginia or Oklahoma are. But ending oil production here would still pose a major challenge for at least one pocket of the Golden State.

About a two-hour drive north of Los Angeles is Kern County, our very own oil country. Kern County produces 70 percent of the state's oil. The industry there is responsible for 16,000 jobs.

And, most crucially, oil and gas generate nearly one-quarter of the county's property tax revenue, which pays for schools, law enforcement, hospitals and other public services. "Nowhere else in California is tied to oil and gas the way we are, and we can't replace what that brings overnight," the county's chief administrative officer, Ryan Alsop, told The New York Times.

My colleague Brad Plumer recently traveled to Kern County and wrote about its struggle to quit oil drilling, even in a state so dedicated to reducing its carbon footprint. Kern County offers a glimpse into issues that other communities across the U.S. could face as they also try to disentangle themselves from fossil fuel income.

The problem boils down to this: If California ends oil drilling, Kern County needs another source of revenue to keep its roads paved and libraries open. Some have suggested leaning into renewables, as the county is already the state's largest provider of wind and solar power. Local leaders are also discussing expanding industries like aerospace, manufacturing, hydrogen or biodiesel production, or even carbon capture technology.

"Politicians, businesses, environmentalists, academics — everyone's sort of thinking hard about how can the county reinvent itself," Brad told me. "But all of these strategies take time, and they might take longer than the county has if the state is serious about phasing out oil production so quickly."

Edit details for each of the following primary sources.  
Add details for any primary sources cited in the text that are not included in this list

☐ 1. Ryan Alsop (Kern County)

Edit Details

☐ 2. Brad Plumer

Edit Details

☐ 3. Nevada County Sheriff's Office

Edit Details

Add New Sources

Brad Plumer

Description: Plumer provides perspective on the efforts in Kern County to find new paths for economic development and the complexities of transitioning away from fossil fuels.

Person Name

Brad Plumer

Person Title

Journalist

Organization

N/A

Document

N/A

Category

media/author

Is this primary source cited in the text?

☐ Yes. I have corrected details where necessary.

☐ Yes, but it's redundant/duplicate of another source.

☐ No

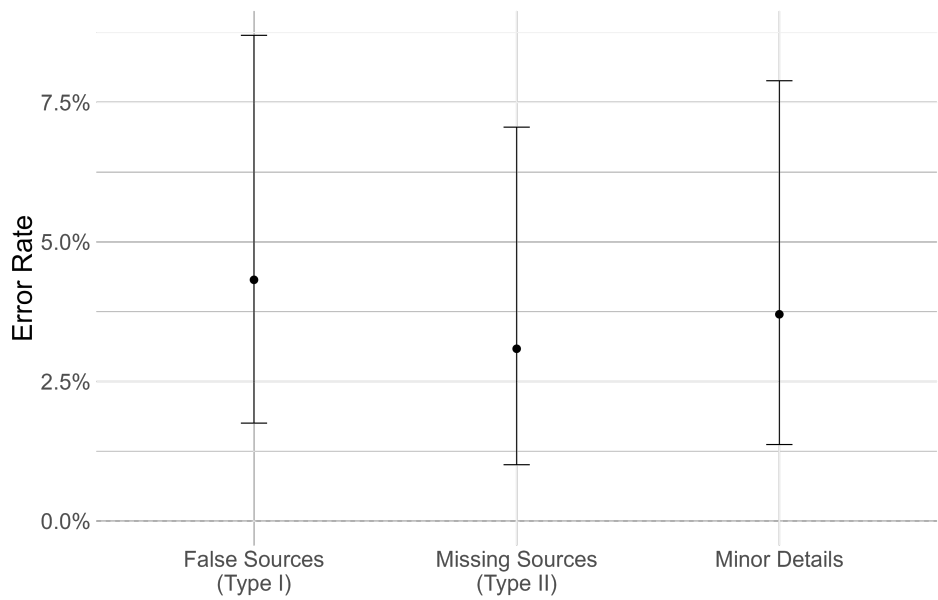
☐ Unsure

Our validation strategy is vulnerable to artificial inflation by inattentive crowdworkers. In the main text, we report data based on results that remove data from inattentive crowd workers, resulting in a smaller validation set of 241 sources. In order to determine crowd worker attentiveness, we asked each crowd worker to correct errors for a “test” article that contained manufactured errors known to us in advance. This test article was embedded in the task, making it impossible for the crowd worker to determine whether there was a test article or which article it was. “Inattentive” workers were those who did not catch these manufactured errors. Thus, we analyze corrections made by eight attentive crowd workers, who each read two articles and made corrections on a total of 214 sources extracted by the GPT-based system.

We identify three types of errors: minor details (i.e., incorrect title, name, or organization); false sources (Type I) where the extracted source was not cited in the article; and missing sources (Type II) where a source present in the article was not extracted. We manually review each error identified by the crowd workers and estimate the overall error rates. Our results show that the GPT-based system is highly accurate in extracting source details and rarely makes Type I or Type II errors (all error rates are less than 5%). Figure A5 shows the error rates with 95% confidence

intervals. Furthermore, a manual inspection shows that the vast majority of the errors are edge cases, where it is difficult to determine with certainty the difference between a source citation and a mere mention of a particular entity (e.g., “President Xi Jinping of China has vowed repeatedly to move ahead with steps in his country to curb climate-altering pollution...”). In particular, when crowd workers noticed that source entities extracted by GPT were not cited in the article (Type I errors), these entities were always at least mentioned in the text. In other words, these errors were exclusively mistakes in judging whether a mentioned entity (i.e., Xi Jinping in the example above) should be considered a cited source, as opposed to outright hallucination of source entities. The remaining true missing source (Type II) errors tend to occur in longer articles with many (6+) sources.

Figure A5: Performance of GPT-based Source Extraction



## C Identifying the Ideological Slant of Sources

**Matching Procedure** We match unstructured organization names from news articles to DIME’s database of 2 million organizations. The main challenge is name variation (e.g., “AARP,” “American Association of Retired People”) and duplication across datasets. To address this, we treat matching as a classification problem, iteratively applying increasingly fine-grained and computationally intensive methods.

First, we compute Jaro-Winkler string distance (via `rapidfuzz`) and retain the top 100 closest DIME names for each source, resulting in 2 million candidate pairs. Next, we calculate weighted Jaccard similarity and cosine similarity for these pairs—metrics that capture different dimensions of textual similarity.

We draw a stratified random sample of 811 pairs, hand-label them as correct or incorrect matches, and use this to train a logistic regression model on the three similarity metrics. The model achieves a test AUC of 0.84. Applying the model, we reject all pairs with a match probability  $p < 0.05$ , eliminating 1.9 million pairs and leaving 100,000 candidates.

**LLM Adjudication** We further refine the matches using GPT-4o with the prompt: “The user will provide the names of two organizations. Explain their relation to each other. Then, categorize their relation as either match or not related. Organizations match if they are the same organization or branches of the same parent organization.” Using the 811 manually labeled cases as a validation set, we find that this prompt has good performance: accuracy 0.92, precision 0.78, recall 0.89, F1 0.83.

We use this prompt to label 10,000 randomly sampled pairs from the dataset of 100,000 remaining pairs. We fine-tune a sentence transformer (all-MiniLM-L6-v2) with these labels. The sentence transformer represents each organization name as a vector in 384-dimensional space. The fine-tuning process ensures the model matching pairs similarly within this vector space. The performance of the fine-tuned sentence transformer is good, although not as good as GPT-4o (based on the test set AUC 0.92, F1 0.75). Using a similar approach to the earlier logistic regression model, we use the 811 manually labeled cases to identify the match with the lowest sentence-transformer cosine similarity ( $s = 0.4$ ). We then reject all candidate pairs with a similarity below this threshold, leaving 40,000 matches remaining. For the remaining matches, we classify them directly using the gpt-4o prompt.

As a final validation exercise we randomly sample 60 pairs, 30 predicted matches and 30 predicted non-matches, sampling 10 non-matches from each stage (10 non-matches that were rejected by the logistic regression, 10 non-matches that were rejected by the sentence transformer, and 10 non-matches that were rejected by gpt-4o). All 60 pairs passed manual inspection (zero errors).

**Ideology Imputation** For sources unmatched to DIME, we impute ideology scores via LLM-based pairwise comparisons (Wu et al., 2023). Using nine ideological “anchor” organizations (ranging from Greenpeace to the Institute for Energy Research), we iteratively prompt GPT-4o to determine each organization’s relative position. For example, Patagonia is placed between the Nature Conservancy and National Wildlife Federation.

The resulting placement is interpolated onto the DIME scale. We impute ideology scores only for sources where such a ranking is meaningful (excluding, for example, universities and bureau-

cratic agencies). Validation shows a correlation of 0.69 between imputed and true DIME scores across all matched organizations, rising to 0.76 for high-quality matches (those with low variance across multiple DIME records).

## **D Journalist Demographics**

We collect information on journalists covering climate news by extracting names from our dataset ( $N = 4,074$ ) and matching them to online profiles. Using automated web searches of LinkedIn and MuckRack—a database and networking platform for journalists—we identify profiles that match both the journalist’s name and their affiliated newspaper. We assume accuracy for MuckRack profiles that match on both name and newspaper and for LinkedIn profiles that match on name and show newspaper-related work experience. A manual validation of 100 randomly matched profiles confirmed 100% accuracy. For each matched profile, we gather work history and educational background. For profiles not found through this process or for incomplete profiles, we manually collected additional journalist information ( $N = 804$ ). In total, we compiled information on 1,842 journalists, responsible for 10,889 articles (83% of the dataset).

For each journalist, we create indicator variables for attending an Ivy League institution (plus Stanford and MIT), holding an advanced (postgraduate) degree, and having studied journalism. We also collect graduation years and earliest recorded work experience to estimate journalist age, and infer gender and race based on names using the `gender` and `wru` R packages. Figure A6 illustrates key trends in journalist demographics over time. The share of younger, female, non-white, and highly educated journalists has risen significantly. Since 2016, the proportion of articles written by journalists under the median age of 37 has nearly doubled, increasing from 10% to 20%. The share of journalists with postgraduate degrees has also grown substantially, from 50% in 2016 to nearly 70% by 2022. Changes in gender and racial composition follow similar patterns but emerge later, around 2018 and 2020, respectively.

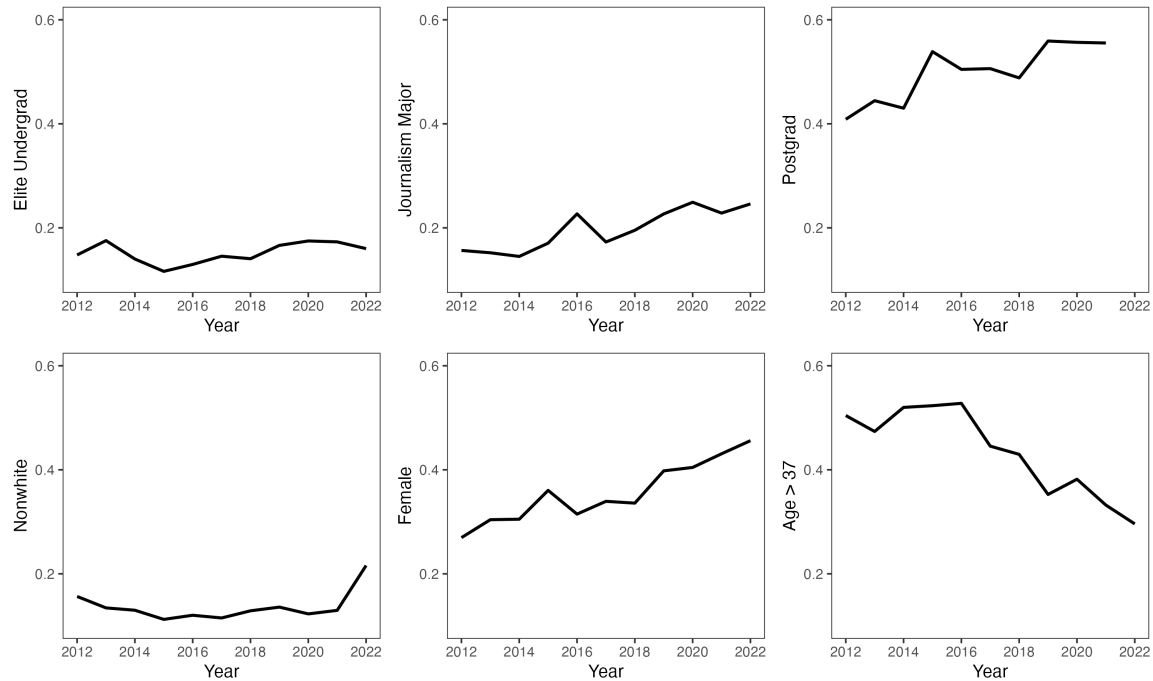
## **E Cohort Effect Additional Analysis**

### **Section fixed effects as a beat assignment proxy**

We use newspaper section as a proxy for journalist beat assignment. Section labels are standardized from raw metadata into 18 categories (e.g., climate, politics, business, science, metro); Table A1 reports the distribution. The largest category, “news,” comprises articles from outlets whose section metadata does not encode a named desk (primarily NYT A-section, Washington Post A-section, and USA Today). Because beat assignment within this category is unobservable, we estimate two robustness specifications that address the concern that unobserved beat variation confounds the cohort estimate.

The first drops all “news”-coded articles, restricting the sample to sections with unambiguous beat-section correspondence. The second replaces the additive outlet and section fixed effects with a fully interacted outlet  $\times$  section fixed effect, absorbing any outlet-specific variation in how beats map to sections. Tables A2 and ?? report results alongside the baseline section FE specification from the main text. Cohort estimates are stable across all three specifications, consistent with beat

Figure A6: Journalist-Level Characteristics Over Time



*Notes:* Shares of journalists who write on climate policy in each year older than the median age (37), with Ivy League undergraduate degree, with journalism degree, female, with postgraduate degree, and non-white predicted race. Journalists in our sample are increasingly younger, with more advanced degrees, female, and non-white over time.

assignment explaining little of the effect.

## Access as alternative explanation

Another alternative explanation for our main result is access to a wider diversity of sources (eg, to economic-cost sources) that arises as a result of journalist seniority. As journalists advance in their careers, they may encounter a wider array of sources, enabling them to sample from a more diverse pool of perspectives. This argument threatens our interpretation especially if urgency-oriented sources are more easy to find and cite for less experienced journalists.

Our descriptive results show that economic-cost arguments are plentiful in political debate in Congressional floor speeches. However, we also can address this concern directly by modeling the prevalence of different source types within journalist over time. We construct a panel dataset at the journalist-year-outlet level counting the total number of urgency, skeptic, economic-cost, and economic-benefit sources cited by each journalist (Results are robust to constructing rates as an article-weighted average rather than pooling source counts). Then, for  $N=1,268$  journalists that appear in at least two years in the data, we analyze whether the relative rate of each source citation changes over time within-journalist. This analysis uses a regression that contains a fixed effect for each journalist, and with tenure (panel year minus first year of experience for each journalist) as the primary independent variable. Thus, the analysis indicates to what extent individual journalists on average change their allocation of sources as they advance in seniority.

Table A1: Article sections

| Section     | Articles | %     |
|-------------|----------|-------|
| news        | 8,668    | 31.7% |
| politics    | 3,107    | 11.4% |
| (missing)   | 2,984    | 10.9% |
| metro       | 2,557    | 9.4%  |
| business    | 2,492    | 9.1%  |
| climate     | 2,207    | 8.1%  |
| world       | 1,351    | 4.9%  |
| arts        | 1,340    | 4.9%  |
| science     | 1,114    | 4.1%  |
| opinion     | 397      | 1.5%  |
| health      | 310      | 1.1%  |
| blog        | 271      | 1.0%  |
| technology  | 194      | 0.7%  |
| sports      | 97       | 0.4%  |
| other       | 67       | 0.2%  |
| real estate | 65       | 0.2%  |
| weather     | 64       | 0.2%  |
| education   | 53       | 0.2%  |

The results are presented in table A3. As journalists advance in tenure, there is no substantive or significant increase in the average rate of urgency citations. There is a significant decrease in the rate of non-urgency citations, suggesting that if anything, increasing journalist experience decreases rates of economy related source citations rather than increasing it.

Table A2: Section FE robustness

| Dependent Variables: |                | Urgency          |                            | Skeptic            |                   |                                 |
|----------------------|----------------|------------------|----------------------------|--------------------|-------------------|---------------------------------|
|                      | + Section FE   | Drop news        | Outlet $\times$ Section FE | + Section FE       | Drop news         | Outlet $\times$ Section FE      |
| Model:               | (1)            | (2)              | (3)                        | (4)                | (5)               | (6)                             |
| Post-2016 cohort     | 0.06<br>(0.04) | 0.07**<br>(0.03) | 0.06<br>(0.04)             | -0.0006<br>(0.002) | -0.002<br>(0.002) | $7.8 \times 10^{-5}$<br>(0.002) |
| <i>Fixed-effects</i> |                |                  |                            |                    |                   |                                 |
| Section              | Yes            | Yes              |                            | Yes                | Yes               |                                 |
| Outlet               | Yes            | Yes              |                            | Yes                | Yes               |                                 |
| Year                 | Yes            | Yes              | Yes                        | Yes                | Yes               | Yes                             |
| source-section_clean |                |                  | Yes                        |                    |                   | Yes                             |
| Observations         | 20,903         | 13,110           | 20,903                     | 20,903             | 13,110            | 20,903                          |

*Clustered (clean\_name) standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

| Dependent Variables: |                   | Econ-Pro          |                            | Econ-Anti           |                     |                            |
|----------------------|-------------------|-------------------|----------------------------|---------------------|---------------------|----------------------------|
|                      | + Section FE      | Drop news         | Outlet $\times$ Section FE | + Section FE        | Drop news           | Outlet $\times$ Section FE |
| Model:               | (1)               | (2)               | (3)                        | (4)                 | (5)                 | (6)                        |
| Post-2016 cohort     | -0.005<br>(0.005) | 0.0002<br>(0.006) | -0.006<br>(0.005)          | -0.01***<br>(0.005) | -0.02***<br>(0.005) | -0.01***<br>(0.005)        |
| <i>Fixed-effects</i> |                   |                   |                            |                     |                     |                            |
| Section              | Yes               | Yes               |                            | Yes                 | Yes                 |                            |
| Outlet               | Yes               | Yes               |                            | Yes                 | Yes                 |                            |
| Year                 | Yes               | Yes               | Yes                        | Yes                 | Yes                 | Yes                        |
| source-section_clean |                   |                   | Yes                        |                     |                     | Yes                        |
| Observations         | 20,903            | 13,110            | 20,903                     | 20,903              | 13,110              | 20,903                     |

*Clustered (clean\_name) standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

*Notes:* “+ Section FE” is the third column from Table ?? in the main text. “Drop news” excludes articles whose section metadata encodes no named desk. “Outlet  $\times$  Section FE” replaces the additive outlet and section fixed effects with their interaction. Standard errors clustered by journalist.

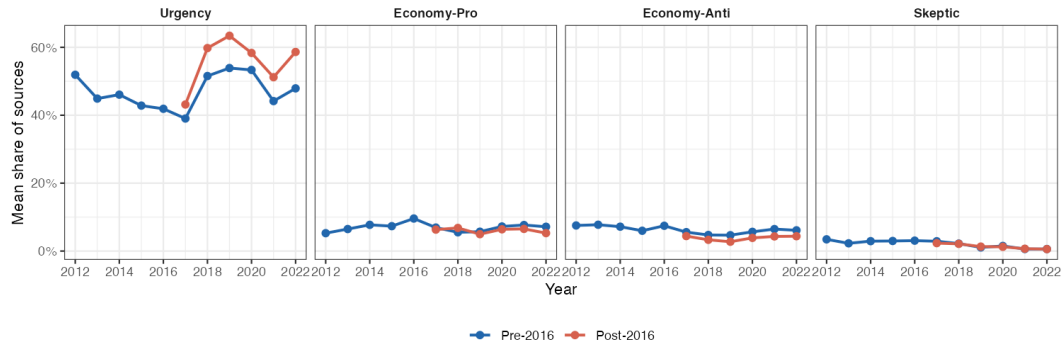
Table A3: Source citation rates by journalist tenure, estimated within-journalist

| Dependent Variables:<br>Model:                               | Urgency<br>(1)   | Econ-Pro<br>(2)     | Econ-Anti<br>(3)    | Skeptic<br>(4)        |
|--------------------------------------------------------------|------------------|---------------------|---------------------|-----------------------|
| tenure                                                       | 0.001<br>(0.003) | -0.003**<br>(0.001) | -0.002**<br>(0.001) | -0.002***<br>(0.0006) |
| Mean DV                                                      | 0.506            | 0.069               | 0.056               | 0.015                 |
| <i>Fixed-effects</i>                                         |                  |                     |                     |                       |
| clean_name                                                   | Yes              | Yes                 | Yes                 | Yes                   |
| Observations                                                 | 4,283            | 4,283               | 4,283               | 4,283                 |
| <i>Clustered (clean_name) standard-errors in parentheses</i> |                  |                     |                     |                       |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>            |                  |                     |                     |                       |



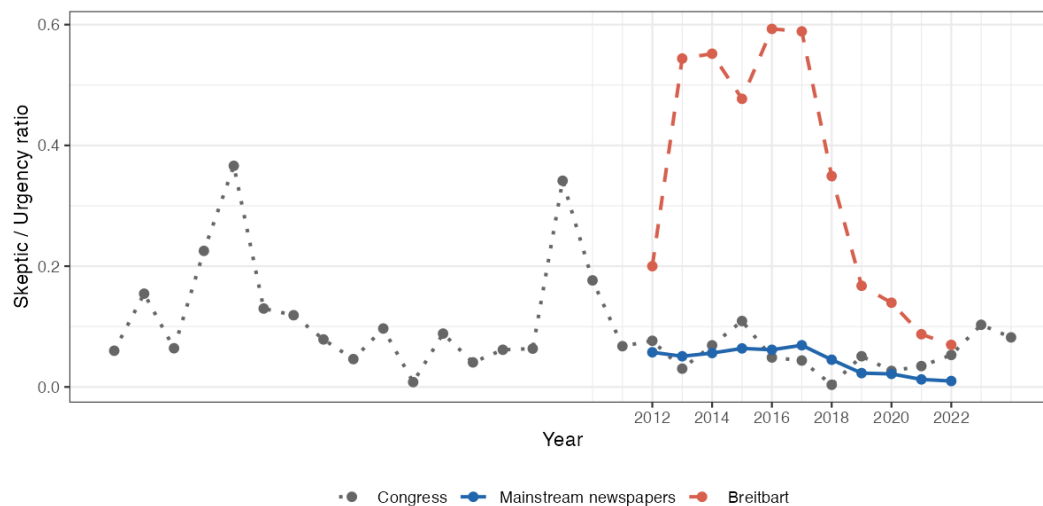
## F Additional Descriptive Figures

Figure A7: Framing Shares Over Time by Pre/Post-2016 Journalist Cohort



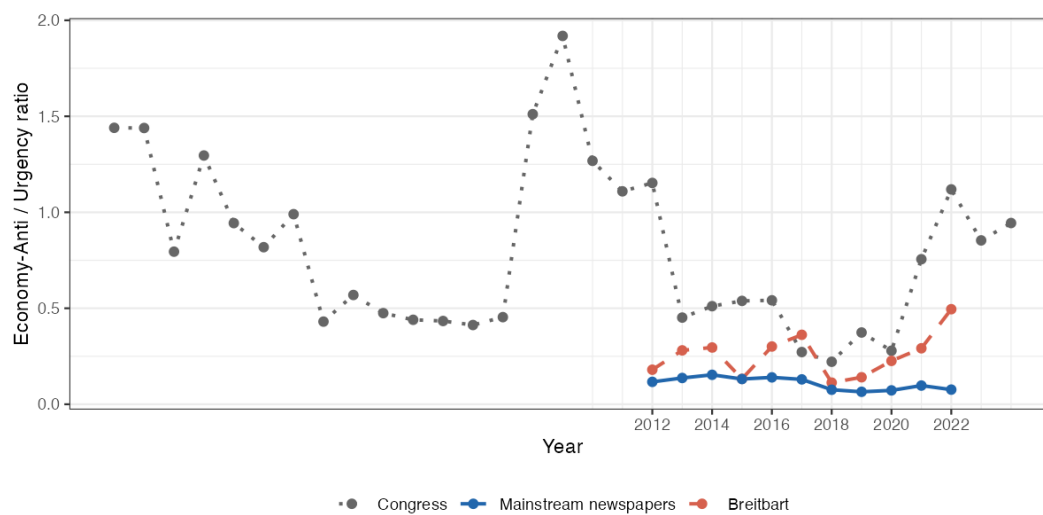
*Notes:* Mean article-level frame shares by year, split by whether the journalist first appears in the data before or after 2016. Mainstream outlets only (Breitbart excluded).

Figure A8: Skeptic / Urgency Ratio: Congress, Mainstream Newspapers, and Breitbart



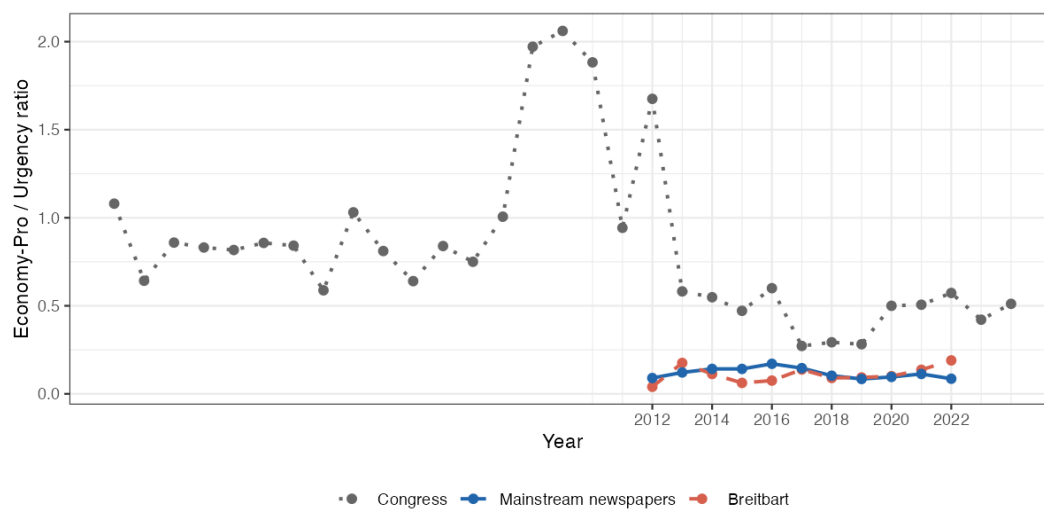
*Notes:* Ratio of skeptic to urgency contributions by year. For Congress, counts of climate speeches; for newspapers, counts of cited sources. Unlike the urgency/economic-cost ratio (Figure 4), this ratio declines similarly across all three information environments, consistent with a general retreat from climate skepticism rather than selective filtering.

Figure A9: economic-cost / Urgency Ratio: Congress, Mainstream Newspapers, and Breitbart



*Notes:* Inverse of Figure 4. Values near zero indicate urgency predominates; values near one indicate parity.

Figure A10: economic-benefit / Urgency Ratio: Congress, Mainstream Newspapers, and Breitbart



*Notes:* Ratio of economic-benefit to urgency contributions by year. For Congress, counts of climate speeches; for newspapers, counts of cited sources.

## **G Cooperative Election Study Measures of Climate Policy Attitudes**

To examine public opinion regarding the trade-off between environmental protection and economic costs, we use the Cooperative Election Study (CES) surveys from 2008–2024. Because the CES questionnaire changed over time, we harmonize responses across survey waves to construct a common measure of support for environmental protection despite potential economic costs.

### **G.1 2008–2012: Jobs versus the Environment**

From 2008 to 2012, respondents were asked:

*Some people think it is important to protect the environment even if it costs some jobs or otherwise reduces our standard of living. Other people think that protecting the environment is not as important as maintaining jobs and our standard of living. Which is closer to the way you feel?*

The response categories were:

1. Much more important to protect the environment even if we lose jobs and have a lower standard of living;
2. Environment somewhat more important;
3. About the same;
4. Economy somewhat more important;
5. Much more important to protect jobs even if the environment worsens.

We code respondents selecting response categories (1) or (2) as prioritizing environmental protection (= 1) and respondents selecting categories (3)–(5) as not prioritizing environmental protection (= 0). Respondents with missing or skipped responses are excluded.

### **G.2 2015–2024: Environmental Policy Proposal**

Beginning in 2015, the CES replaced the jobs-versus-environment question with a battery of environmental policy proposals. To maintain comparability over time, we use the proposal that most closely captures respondents' willingness to incur economic costs for environmental protection:

*Strengthen the Environmental Protection Agency's enforcement of the Clean Air Act and Clean Water Act even if it costs U.S. jobs.*

Respondents were asked whether they *support* or *oppose* this proposal. We code respondents who support the proposal as prioritizing environmental protection (= 1) and those who oppose it as (= 0). Respondents with missing or skipped responses are excluded.

Although the wording changes after 2012, both survey instruments explicitly ask respondents to weigh environmental protection against potential economic costs, making them comparable measures of support for climate policy.